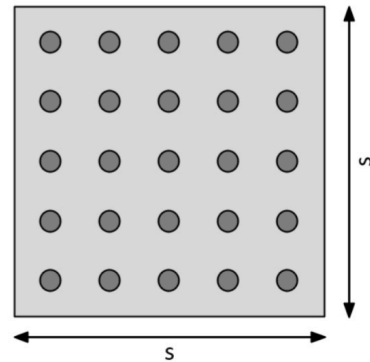
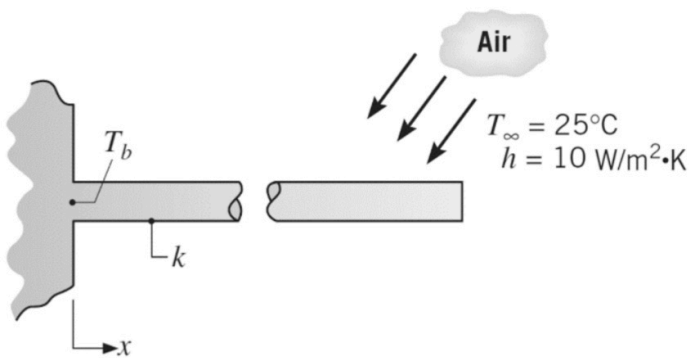


Problem 1

Known: $r = 1 \text{ mm}$, $L = 5 \text{ cm}$, $T_\infty = 25^\circ\text{C}$, $T_b = 125^\circ\text{C}$,
 $h = 10 \frac{\text{W}}{\text{m}^2\text{K}}$, $k = 200 \frac{\text{W}}{\text{mK}}$, $s = 3 \text{ cm}$, $q'' = 1000 \frac{\text{W}}{\text{m}^2}$,
 $L_1 = 2 \text{ cm}$, $k_1 = 10 \frac{\text{W}}{\text{mK}}$, $R_c'' = 10^{-4} \frac{\text{m}^2\text{K}}{\text{W}}$

- find: a) Fin temp @ $x = \frac{L}{2}$, $x = L$
 b) ϵ_f , η_f
 c) η_f using table 3.5
 d) R_f of single fin
 e) for 5×5 array: η_o , R_o , q_{inf} , q_{fins}
 f) Thermal resistance network, T_h , compare to w/o heat sink (temp reduction)

Schematic:



assumptions: steady state, 1D conduction, constant k & h , no q ,
 base is isothermal

analysis:

a) Table 3.4: $\frac{\Theta}{\Theta_b} = \frac{\cosh[m(L-x)]}{\cosh(mL)}$

$$T(x) = T_\infty + \Theta_b \frac{\cosh[m(L-x)]}{\cosh(mL)}$$

$$T(L/2) = 25 + 100 \frac{\cosh[10(0.05 - 0.025)]}{\cosh(10 \cdot 0.05)}$$

$$\left\{ \begin{array}{l} \Theta_b = 125 - 25 \\ A_c = \pi(0.001)^2 \\ P = 2\pi(0.001) \\ m = \sqrt{\frac{2(10)}{200(0.001)}} = 10 \text{ m}^{-1} \end{array} \right.$$

Problem 1 cont'd

$$T(L/2) = 25 + 91.47 = 116.47^\circ\text{C}$$

$$T(L) = 25 + 100 \frac{\cosh[10(.05 - .05)]}{\cosh(.5)}$$

$$T(L) = 25 + 88.68 = 113.68^\circ\text{C}$$

$$T(L/2) = 116.47^\circ\text{C}$$

$$T(L) = 113.68^\circ\text{C}$$

$$b) q_f = m k A_c \theta_b \tanh(mL)$$

$$= (10)(200)(3.14 \times 10^{-6})(100)(.4621) = 0.29 \text{ W}$$

$$n_f = \frac{q_f}{h A_f \theta_b}, \quad A_f = PL = (.066283)(.05) = 3.14 \times 10^{-4} \text{ m}^2$$

$$n_f = \frac{.2904}{(10)(3.14 \times 10^{-4})(100)} \Rightarrow n_f = 0.924$$

$$\varepsilon_f = \frac{q_f}{h A_c \theta_b} = \frac{.2904}{(10)(3.14 \times 10^{-6})(100)} \Rightarrow \varepsilon_f = 92.42$$

$$c) L_c = L + \frac{D}{4} = .05 + \frac{.002}{4} = .0505 \text{ m}$$

$$n_f = \frac{\tanh(mL_c)}{mL_c} = \frac{\tanh(10 \times .0505)}{10 \times .0505}$$

$$n_f = 0.923$$

Table 3.5 gives almost identical answer.

It is a good approximation.

Problem 1 cont'd

$$d) R_f = \frac{\theta_b}{q_f} = \frac{100}{.2904} \Rightarrow R_f = 344.4 \frac{K}{W}$$

$$e) \# \text{ fins} = 25$$

$$A_{\text{plate}} = 9 \times 10^{-4} \text{ m}^2$$

$$A_b = A_{\text{plate}} - NA_c = 9 \times 10^{-4} - 25(3.14 \times 10^{-6}) = 8.21 \times 10^{-4} \text{ m}^2$$

$$NA_f = 25(3.14 \times 10^{-4}) = 7.854 \times 10^{-3} \text{ m}^2$$

$$A_t = A_b + NA_f = 8.21 \times 10^{-4} + 7.854 \times 10^{-3} = 8.675 \times 10^{-3} \text{ m}^2$$

$$\eta_o = 1 - \frac{NA_f}{A_t} (1 - \eta_f) = 1 - \frac{7.854 \times 10^{-3}}{8.6754 \times 10^{-3}} (1 - .924)$$

$$\eta_o = 0.931$$

$$R_o = \frac{1}{\eta_o h A_t} = \frac{1}{(.931)(10)(8.6754 \times 10^{-3})}$$

$$R_o = 12.38 \frac{K}{W}$$

$$q_{\text{unf}} = h A_b \theta_b = (10)(8.21 \times 10^{-4})(100)$$

$$q_{\text{unf}} = 0.821 \text{ W}$$

$$q_{\text{fins}} = N q_f = 25(.2904)$$

$$q_{\text{fins}} = 7.26 \text{ W}$$

Problem 1 cont'd

f) Resistance network:



$$R_c = \frac{R_c''}{A}, \quad R_{cond,1} = \frac{L_1}{k_1 A}, \quad R_o = \frac{1}{\eta_o h A_t}$$

$$R_{tot,hs} = R_c + R_{cond,1} + R_o$$

$$\dot{Q} = q'' A = (1000)(9.0 \times 10^{-4}) = 0.9 \text{ W}$$

$$R_c = \frac{10^{-4}}{9 \times 10^{-4}} = 0.11 \text{ K/W}$$

$$R_{cond,1} = \frac{L_1}{k_1 A} = \frac{0.02}{(10)(9 \times 10^{-4})} = \frac{.02}{.009} = 2.22 \text{ K/W}$$

$$R_{tot,hs} = 0.11 + 2.22 + 12.38 = 14.71 \text{ K/W}$$

$$T_h = T_\infty + \dot{Q} R_{tot,hs} = 25 + (.9)(14.71)$$

$$T_h = 38.2^\circ \text{C}$$

w/o heat sink:

$$R_{conv} = \frac{1}{(10)(9 \times 10^{-4})} = 111.11 \text{ K/W}$$

$$R_{tot,nohs} = R_c + R_{cond,1} + R_{conv} = 0.11 + 2.22 + 111.11 = 113.44 \text{ K/W}$$

$$T_{h,nohs} = T_\infty + \dot{Q} R_{tot,nohs} = 25 + (0.9)(113.44) = 127.1^\circ \text{C}$$

$$\Delta T = 127.1 - 38.2 \Rightarrow \Delta T = 88.9^\circ \text{C}$$